

The CoM offset under ground effect, channel by channel

companion to *cosh.methodology*; gains (α_f, α_M) from Table 3 at $\varphi^* = 0$ (interference model, $\alpha_f = \alpha_M = 4.31\%$)

1 The question

The delivered CoM offset is read from the static tip-over balance. If that balance is written *without* ground effect while the vehicle actually tips *with* it, the identified offset differs from the true one. This note derives the difference in closed form, separates it into its two channels, and evaluates both on the campaign. The punchline: per tip direction the error is millimetres, but the deliverable's direction pairing cancels the dominant channel identically, leaving at most 0.67 mm — a 4.31% multiplicative scale factor plus a small geometric asymmetry.

2 The two balances

Write the onset balance per tip direction $s = \pm 1$ in the matched-thrust convention (companion document), with l_s the direction's pivot arm, f_s the collective thrust at onset, and $S_{\text{off}} = W \times$ (signed offset):

$$(1 + \alpha_M) M_s + s \alpha_f f_s l_s = s (W - f_s) l_s + S_{\text{off}}. \quad (1)$$

The no-GE model reads the offset from the same data by solving

$$S_{\text{read},s} = M_s - s (W - f_s) l_s. \quad (2)$$

Substituting (1) into (2) gives the per-direction reading error in one line:

$$\boxed{S_{\text{read},s} = S_{\text{off}} - \alpha_M M_s - s \alpha_f f_s l_s} \quad (3)$$

3 One direction alone: millimetres

Both channels of (3) are large per direction: $\alpha_M M_s \approx 0.043 \times 1\text{--}1.8 \text{ N m} \approx 40\text{--}80 \text{ mN m}$, and $\alpha_f f_s l_s \approx 0.043 \times 21 \times 0.11\text{--}0.14 \approx 100\text{--}130 \text{ mN m}$. Divided by $W \approx 31 \text{ N}$, a single-direction offset reading is off by

$$\text{median } 5.2 \text{ mm}, \quad \text{max } 6.5 \text{ mm}$$

over the campaign — comparable to the offsets being identified. This is the quantitative reason the deliverable never uses one direction alone.

4 The direction pair: what cancels, what survives

The deliverable averages the two directions, $S_{\text{read}} = \frac{1}{2}(S_{\text{read},+} + S_{\text{read},-})$. In (3) the thrust-channel term carries the factor s , so its *symmetric* part — the $\pm 5 \text{ mm}$ monster of the previous section — cancels identically, and

$$\boxed{S_{\text{off}} - S_{\text{read}} = \alpha_M M_{\text{ff}} + \frac{\alpha_f}{2}(f_+ l_+ - f_- l_-)}, \quad M_{\text{ff}} = \frac{1}{2}(M_+ + M_-), \quad (4)$$

or in millimetres of offset, dividing by W . Two channels remain:

The moment channel $\alpha_M M_{\text{ff}}/W$. Since $M_{\text{ff}} \approx W \times$ offset up to the arm asymmetry, this term is α_M of the offset itself: the no-GE reading under-estimates the offset magnitude by the factor $1/(1 + \alpha_M)$ — a multiplicative 4.31% scale error that no pairing can remove, because it multiplies the very quantity being delivered.

The thrust channel $\frac{\alpha_f}{2}(f_+ l_+ - f_- l_-)/W$. Only the direction *asymmetry* of fl survives the pair; with symmetric thrust and arms it vanishes exactly.

5 The campaign numbers

Evaluated per configuration from the identified moments, onset thrusts and odometry-fitted arms (`mcrit_prediction.csv`):

config	M_{ff} [N m]	moment ch. [mm]	thrust ch. [mm]	total [mm]
case_01/Mx	−0.048	−0.069	−0.098	−0.167
case_01/My	+0.336	+0.482	+0.154	+0.636
case_02/Mx	−0.479	−0.653	−0.006	−0.659
case_02/My	+0.334	+0.456	+0.188	+0.644
case_03/Mx	−0.272	−0.371	−0.059	−0.430
case_03/My	−0.023	−0.031	+0.124	+0.093
case_04/Mx	+0.165	+0.226	−0.041	+0.185
case_04/My	−0.007	−0.010	+0.168	+0.158
case_05/Mx	+0.341	+0.465	−0.070	+0.395
case_05/My	+0.334	+0.455	+0.214	+0.669
median		0.41	0.11	≈ 0.4
max		0.65	0.21	0.67

The moment channel dominates and tracks M_{ff} exactly as (4) says; the thrust channel is the small residue of the *fl* asymmetry. For scale, the load-cell validation RMS of the delivered offsets is 1.64 mm: the entire ground-effect footprint on the deliverable is below half of it.

6 Why the correction is reported, not applied

(4) says the true offset is the read one *plus* up to 0.67 mm; the honest question is whether adding it improves the agreement with the load-cell truth. It does not (`analysis/ge_offset_effect.py`): the corrected offsets sit slightly *farther* from truth, by an amount inside the scatter. The reason is a degeneracy the static data cannot break — a common shift of the contact resultant by a few millimetres produces the same signature, and the two mechanisms cannot be separated without a second ground clearance. The correction is therefore reported as a *bound on the channel*: the delivered offset is insensitive to the static ground effect at the ≤ 0.7 mm level, dominated by the multiplicative 4.31% of the moment channel.

Reproduction

```
python analysis/mcrit_prediction.py <outdir>    # writes the CSVs
python analysis/ge_offset_effect.py <outdir>    # the channel split
```

The table above is (4) evaluated on `mcrit_prediction.csv` with $\alpha_f = \alpha_M = 0.0431$; the single-direction numbers are (3) evaluated per row.